

HALIREENA RUSHDIHA MOHOMED

Bioinformatician | Crop Transcriptomics | UAV Phenotyping | Reproducible Omics Analysis
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PROFILE

MSc Bioinformatics student at the University of Birmingham Dubai (81% taught average, graduating November 2026) with First Class Honours in Biotechnology. Through an internship, my thesis and coursework I have built reproducible R pipelines for RNA-seq, microarray and UAV-image data, validating every result before reporting it. I have presented my research at an international conference and led pipeline validation in a team that won first place in a genomics sprint. I am looking for a graduate bioinformatics role in plant science or agritech where I can keep building my analysis and programming skills.

KEY SKILLS

Programming: R; Python; Bash; R Markdown; Git/GitHub

Analysis: R packages (limma, DESeq2, clusterProfiler, ggplot2); RNA-seq (FastQC, MultiQC, cutadapt, Bowtie2, StringTie, featureCounts, DESeq2); microarray analysis; GO and KEGG enrichment; CNV analysis; Random Forest; UAV multispectral image analysis

Platforms: BlueBEAR HPC (SLURM job arrays); Galaxy; MetaboAnalyst; GEO, ArrayExpress, Ensembl, Phytosome, NCBI, KEGG, BLAST, MEGA

Laboratory: DNA extraction, PCR, gel electrophoresis, spectrophotometry, hydroponic plant culture

EDUCATION

University of Birmingham Dubai

Sep 2025 – Nov 2026 (expected)

MSc Bioinformatics

- Taught-module average 81% (Distinction level); awarded a 50% University of Birmingham 125th Anniversary Scholarship for academic merit.
- Key modules: Introduction to Biology and Programming 98%; Genomics and Next Generation Sequencing 86%; Mathematics and Statistics 84%.

Northumbria University, at BMS School of Science, Colombo

Jan 2024 – Nov 2024

BSc (Hons) Biotechnology, First Class Honours (79%)

- Dissertation 84% (see Research Projects); Dean's List Award; BMS Best Student Award 2024 (70% or more in every module).

BMS School of Science, Colombo

Mar 2022 – Jan 2024

Higher Diploma in Biotechnology, Distinction

- Dissertation: extracted DNA and amplified the *rbcl* barcode gene by PCR in Sri Lankan pumpkin varieties, verifying products by gel electrophoresis.

WORK EXPERIENCE

Bioinformatics Intern | RIMHS, University of Sharjah (with Abiomix)

Feb 2026 – May 2026

- Analysed a public microarray dataset of potato roots under phosphate starvation (ArrayExpress E-MTAB-2990) by writing a five-stage R pipeline: quality control, limma differential expression, clustering and GO enrichment with clusterProfiler.
- Checked that the findings held under two analysis settings (285 vs 227 differentially expressed genes, 184 shared) and identified a clear phosphate-starvation response: all 11 acid phosphatases and all 6 phosphate transporters upregulated.
- Wrote up methods and limitations for the research team, including a KEGG gene-ID incompatibility, and ran a short machine-learning teaching session for fellow interns.

Medical Transcriptionist Trainee | Hemas Hospital, Sri Lanka

Oct 2024 – Dec 2024

- Transcribed 40+ dermatology and ophthalmology consultations a day into accurate clinical records with consultants, nurses and IT staff during the move to electronic health records.

Laboratory Intern | Industrial Technology Institute (ITI), Sri Lanka

Feb 2023

- Carried out DNA extraction, PCR, gel electrophoresis and spectrophotometry to standard protocols in a two-week placement; reviewed results and flagged inconsistencies; awarded Best Performance.

RESEARCH PROJECTS

MSc Thesis | Potato late blight severity detection from UAV RGB and multispectral imagery

Apr 2026 – Sep 2026

Supervisor: Dr Lindsey Compton

- Turned drone imagery of 250 potato varieties into plot-level features and trained Random Forest and other machine-learning models to detect late blight.
- Evaluated the models with cross-validation grouped by variety and permutation tests, and reframed the question with my supervisor to state what aerial imagery can and cannot detect. Presented 18 Sep 2026; abstract and paper in preparation.

BioConnect 2026 Sprint (Abiomix challenge) | Prioritising unsolved rare-disease cases for genomic re-analysis, team of 3

Jul 2026

- With two teammates, built a seven-stage pipeline that ranked 133 unsolved cases for re-analysis; I led the GitHub repository, pipeline validation and CNV analysis, and coordinated tasks and deadlines.
- First place in the Abiomix challenge; selected for the BioConnect Accelerator; poster at the 4th International UAE Rare Disease Society Congress; manuscript in preparation.

BSc Dissertation | Iron biofortification of leafy greens in hydroponic systems

Feb 2024 – Aug 2024

- Grew mukunuwenna (*Alternanthera sessilis*) and kangkung in nutrient-film and coconut-coir hydroponics at three iron levels and assayed iron, nutrient, phenolic and antioxidant content.
- Measured leaf iron by AAS (one-way ANOVA): iron rose with dose in coconut-coir mukunuwenna (to 20.3 ppm) and NFT kangkung (19.5 ppm) but not in NFT mukunuwenna. Published as an abstract and presented orally at ICIET 2024; mark 84%.

PUBLICATIONS AND PRESENTATIONS

- Mohomed HR et al. (2024).** Iron biofortification of *Alternanthera sessilis*: a comparative study of hydroponics techniques. *Proc. ICIET 2024*, University of Sri Jayewardenepura, p. 23, ISSN 2815-0317. Oral presentation. iciet.sjp.ac.lk/.../ICIET-2024-e-Proceedings_2024Nov28-1.pdf
- Baroudi R, Latheef FN, **Mohomed HR**, Waasia FZ, Toure SM, El-Hadidi M (2026). Transparent phenotype-driven variant reanalysis for unsolved rare disease cases in a Middle Eastern cohort. Poster 72, *4th International UAE Rare Disease Society Congress*, Dubai.

LANGUAGES

English (fluent), Sinhala (native), Tamil, Hindi, Sri Lankan Malay (understanding)

Project write-ups, figures and code at halireena.github.io. References available on request.